We create chemistry

Safety Data Sheet
Pasadena Oxo Middle Boiler Cut

1. Identification

Product identifier used on the label

Pasadena Oxo Middle Boiler Cut

Recommended use of the chemical and restriction on use

* The “Recommended use” identified for this product is provided solely to comply with a Federal requirement and is not part of the seller's published specification. The terms of this Safety Data Sheet (SDS) do not create or infer any warranty, express or implied, including by incorporation into or reference in the seller's sales agreement.

Details of the supplier of the safety data sheet

Company:
BASF CORPORATION
100 Park Avenue
Florham Park, NJ 07932, USA

Telephone: +1 973 245-6000

Emergency telephone number

CHEMTREC: 1-800-424-9300
BASF HOTLINE: 1-800-832-HELP (4357)

Other means of identification

Chemical family: Preparation based on: hydrocarbons

2. Hazards Identification

According to Regulation 2012 OSHA Hazard Communication Standard; 29 CFR Part 1910.1200

Classification of the product

Asp. Tox.	1	Aspiration hazard
Flam. Liq.	4	Flammable liquids
Skin Corr./Irrit.	2	Skin corrosion/irritation
Eye Dam./Irrit.	1	Serious eye damage/eye irritation
Skin Sens.	1B	Skin sensitization
Repr.	2 (unborn child)	Reproductive toxicity
STOT RE	1	Specific target organ toxicity — repeated exposure

Label elements

Pictogram:

Signal Word:
Danger

Hazard Statement:

H227	Combustible liquid.
H318	Causes serious eye damage.
H315	Causes skin irritation.
H317	May cause an allergic skin reaction.
H304	May be fatal if swallowed and enters airways.
H361	Suspected of damaging the unborn child.
H372	Causes damage to organs (Central nervous system, Peripheral nervous system) through prolonged or repeated exposure.

Precautionary Statements (Prevention):

P280	Wear protective gloves/protective clothing/eye protection/face protection.
P260	Do not breathe dust/gas/mist/vapours.
P210	Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P201	Obtain special instructions before use.
P202	Do not handle until all safety precautions have been read and understood.
P270	Do not eat, drink or smoke when using this product.
P264	Wash with plenty of water and soap thoroughly after handling.
P272	Contaminated work clothing should not be allowed out of the workplace.

Precautionary Statements (Response):

P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P310	Immediately call a POISON CENTER or doctor/physician.
P303 + P352	IF ON SKIN (or hair): Wash with plenty of soap and water.
P362 + P364	Take off contaminated clothing and wash it before reuse.
P331	Do NOT induce vomiting.
P370 + P378	In case of fire: Use extinguishing powder, foam or CO2 for extinction.

Precautionary Statements (Storage):

P405	Store locked up.
P403	Store in a well-ventilated place.

Precautionary Statements (Disposal):

P501	Dispose of contents/container to hazardous or special waste collection point.

3. Composition / Information on Ingredients

According to Regulation 2012 OSHA Hazard Communication Standard; 29 CFR Part 1910.1200

CAS Number	Weight %	Chemical name
123-05-7	>= 0.0 - < 3.0%	2-ethylhexanal
123-72-8	>= 0.0 - < 3.0%	butyraldehyde

603-35-0	>= 1.0 - < 7.0%	triphenylphosphine
645-62-5	>= 0.0 - < 3.0%	2-ethylhex-2-enal
6846-50-0	>= 5.0 - < 20.0%	Propanoic acid, 2-methyl-, 2,2-dimethyl-1-(1-methylethyl)-1,3- propanediyl ester
68609-68-7	>= 3.0 - < 15.0%	high boiling by-products from the manufacturing process of 2-ethylhexanol

4. First-Aid Measures

Description of first aid measures

General advice:
First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). Immediately remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention. Immediately administer a corticosteroid from a controlled/metered dose inhaler.

If on skin:
Immediately wash thoroughly with plenty of water, apply sterile dressings, consult a skin specialist.

If in eyes:
Immediately wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

If swallowed:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Most important symptoms and effects, both acute and delayed

Symptoms: The most important known symptoms and effects are described in the labelling (see section 2) and/or in section 11.

Indication of any immediate medical attention and special treatment needed

Note to physician
Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Extinguishing media

Suitable extinguishing media:
foam, carbon dioxide, dry powder

Special hazards arising from the substance or mixture

Hazards during fire-fighting:
Vapours may form explosive mixture with air. Cool endangered containers with water-spray.

Advice for fire-fighters

Protective equipment for fire-fighting:
Wear a self-contained breathing apparatus.

Further information:
Fire debris must be disposed of in accordance with offical regulations.

6. Accidental release measures

Personal precautions, protective equipment and emergency procedures
Handle in accordance with good industrial hygiene and safety practice. Avoid contact with the skin, eyes and clothing. Avoid inhalation.

Environmental precautions
Do not discharge into drains/surface waters/groundwater. Discharge into the environment must be avoided.

Methods and material for containment and cleaning up
For large amounts: Pump off product.
For residues: Pick up with suitable absorbent material (e.g. sand, sawdust, general-purpose binder, kieselguhr). Dispose of absorbed material in accordance with regulations.
Dispose of in accordance with national, state and local regulations.

7. Handling and Storage

Precautions for safe handling
Ensure thorough ventilation of stores and work areas.

Protection against fire and explosion:
Prevent electrostatic charge - sources of ignition should be kept well clear - fire extinguishers should be kept handy.

Conditions for safe storage, including any incompatibilities

Suitable materials for containers: Stainless steel 1.4301 (V2), Stainless steel 1.4401, Carbon steel (Iron)

Further information on storage conditions: Containers should be stored tightly sealed in a dry place.

8. Exposure Controls/Personal Protection

No occupational exposure limits known.

Personal protective equipment

Respiratory protection:
When workers are facing concentrations above the occupational exposure limits they must use appropriate certified respirators. Wear a NIOSH-certified (or equivalent) organic vapour/particulate respirator.

Observe OSHA regulations (29 CFR 1910.134-139).

Hand protection:
Wear chemical resistant protective gloves.

Eye protection:
Tightly fitting safety goggles (chemical goggles). Wear face shield if splashing hazard exists.

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. head protection, apron, protective boots, chemical-protection suit.

General safety and hygiene measures:
Avoid contact with the skin, eyes and clothing. Handle in accordance with good industrial hygiene and safety practice.

9. Physical and Chemical Properties

Form:	liquid	
Odour:	of petroleum distillate (e.g. gasoline, kerosene)	
Colour:	pale yellow green, clear	
Boiling range:	96 - 200 °C	
Flash point:	77.5 - 80.5 °C	(ASTM D93)
Vapour pressure:	0.1 - 2 hPa	
Density:	0.9438 - 0.9441 g/cm3 (20 °C)	
Viscosity, dynamic:	18.7 - 19.5 cps (20 °C)	
Viscosity, kinematic:	19.8 cst (20.8 °C)	
Solubility in water:	insoluble	

10. Stability and Reactivity

Reactivity
No hazardous reactions if stored and handled as prescribed/indicated.

Corrosion to metals:
Corrosive effects to metal are not anticipated.

Chemical stability
The product is stable if stored and handled as prescribed/indicated.

Possibility of hazardous reactions
Violent reaction under influence of oxidizing agents.

Conditions to avoid
Avoid all sources of ignition: heat, sparks, open flame. Heating may cause pressure build-up and possible rupture of the container.

Incompatible materials
strong oxidizing agents

Hazardous decomposition products

Decomposition products:

No hazardous decomposition products if stored and handled as prescribed/indicated.

11. Toxicological information

Primary routes of exposure

Routes of entry for solids and liquids are ingestion and inhalation, but may include eye or skin contact. Routes of entry for gases include inhalation and eye contact. Skin contact may be a route of entry for liquefied gases.

Acute Toxicity/Effects

Acute toxicity
Assessment of acute toxicity: Not expected to be acutely toxic.
Virtually nontoxic after a single ingestion. Virtually nontoxic after a single skin contact. Virtually nontoxic by inhalation.

Information on: 2-ethylhexanal
Assessment of acute toxicity:Of low toxicity after single ingestion. Virtually nontoxic after a single skin contact. Virtually nontoxic by inhalation.

Information on: butyraldehyde
Assessment of acute toxicity:Of low toxicity after single ingestion. Virtually nontoxic by inhalation. Of low toxicity after short-term skin contact.

Information on: triphenylphosphine
Assessment of acute toxicity:Of moderate toxicity after single ingestion. Virtually nontoxic after a single skin contact. Virtually nontoxic by inhalation.

Information on: 2-ethylhex-2-enal
Assessment of acute toxicity:Of low toxicity after single ingestion. The inhalation of a highly enriched/saturated vapor-air-mixture represents an unlikely acute hazard.

Information on: Propanoic acid, 2-methyl-, 2,2-dimethyl-1-(1-methylethyl)-1,3- propanediyl ester
Assessment of acute toxicity:Virtually nontoxic after a single skin contact. Of low toxicity after single ingestion. Virtually nontoxic by inhalation.

Oral
Type of value: ATE
Value: > 5,000 mg/kg

Inhalation
Type of value: ATE
Value: > 20 mg/l
Determined for vapor

Type of value: ATE
Value: > 5 mg/l
Determined for mist

Dermal
Type of value: ATE
Value: > 5,000 mg/kg

Irritation / corrosion
Assessment of irritating effects: Causes skin irritation. Causes serious eye damage.

Information on: 2-ethylhexanal
Assessment of irritating effects: Not irritating to the eyes. May cause slight irritation to the skin.

Information on: butyraldehyde
Assessment of irritating effects: Not irritating to the skin. Eye contact causes irritation.

Information on: triphenylphosphine
Assessment of irritating effects: Not irritating to the skin. May cause severe damage to the eyes.

Information on: 2-ethylhex-2-enal
Assessment of irritating effects: Skin contact causes irritation. Not irritating to the eyes.

Information on: Propanoic acid, 2-methyl-, 2,2-dimethyl-1-(1-methylethyl)-1,3- propanediyl ester
Assessment of irritating effects: Skin contact causes irritation.

Sensitization
Assessment of sensitization: Sensitization after skin contact possible.

Information on: 2-ethylhexanal
Assessment of sensitization:
sensitizing effect in animal tests

Information on: triphenylphosphine
Assessment of sensitization:
Caused skin sensitization in animal studies.

Information on: 2-ethylhex-2-enal
Assessment of sensitization:
Caused skin sensitization in animal studies.

Aspiration Hazard
May also damage the lung at swallowing (aspiration hazard).

Chronic Toxicity/Effects

Repeated dose toxicity
Assessment of repeated dose toxicity: Repeated exposure to small quantities may affect certain organs.

Information on: butyraldehyde
Assessment of repeated dose toxicity: The substance may cause damage to the upper respiratory tract after repeated inhalation, as shown in animal studies.

Information on: triphenylphosphine
Assessment of repeated dose toxicity: Repeated exposure may affect certain organs. Damages the central nerve system. Damages the peripheral nerve system.

Information on: 2-ethylhex-2-enal
Assessment of repeated dose toxicity: No data available.

Information on: Propanoic acid, 2-methyl-, 2,2-dimethyl-1-(1-methylethyl)-1,3- propanediyl ester
Assessment of repeated dose toxicity: Repeated exposure to large quantities may affect certain organs.

Genetic toxicity
Assessment of mutagenicity: Based on available Data, the classification criteria are not met.

Carcinogenicity
Assessment of carcinogenicity: Based on available Data, the classification criteria are not met.

Reproductive toxicity
Assessment of reproduction toxicity: Based on available Data, the classification criteria are not met.

Teratogenicity
Assessment of teratogenicity: May cause harm to the unborn child.

Information on: 2-ethylhexanal
Assessment of teratogenicity: Indications of possible developmental toxicity/teratogenicity were seen in animal studies.

Other Information
The product has not been tested. The statement has been derived from the properties of the individual components.

Symptoms of Exposure

The most important known symptoms and effects are described in the labelling (see section 2) and/or in section 11.

12. Ecological Information

Toxicity

Aquatic toxicity
Assessment of aquatic toxicity:
Based on available Data, the classification criteria are not met.

Aquatic toxicity

Information on: 2-ethylhexanal
Assessment of aquatic toxicity:
Acutely toxic for aquatic organisms. Depending on local conditions and existing concentrations, disturbances in the biodegradation process of activated sludge are possible.

Information on: butyraldehyde
Assessment of aquatic toxicity:
Acutely harmful for aquatic organisms.
The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Information on: 2-ethylhex-2-enal
Assessment of aquatic toxicity:
Acutely harmful for aquatic organisms. The chronic aquatic risk classification is based on acute aquatic toxicity study data and the environmental fate properties of the product. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Information on: high boiling by-products from the manufacturing process of 2-ethylhexanol

Assessment of aquatic toxicity:
Acutely harmful for aquatic organisms. Inhibition of degradation activity in activated sludge is not to be anticipated during correct introduction of low concentrations.

Additional information

Other ecotoxicological advice:
The product has not been tested. The statements on ecotoxicology have been derived from the properties of the individual components. The product has been assessed on the basis of the components' available data. To some extent data gaps exist for individual components. According to our present knowledge and experience dangers which are not covered by the current labeling are not to be expected.

13. Disposal considerations

Waste disposal of substance:
Dispose of in accordance with local authority regulations. Disposal requirements are dependent on the hazard classification and will vary by location and the type of disposal selected. All waste materials should be reviewed to determine the applicable hazards (testing may be necessary).

Container disposal:
Dispose of in accordance with national, state and local regulations.

14. Transport Information

Land transport
USDOT

Classified as combustible liquid in containers greater than 119 gallons.

Sea transport
IMDG

Not classified as a dangerous good under transport regulations

Air transport
IATA/ICAO

Not classified as a dangerous good under transport regulations

15. Regulatory Information

Federal Regulations

Registration status:
Chemical TSCA, US released / listed

EPCRA 311/312 (Hazard categories): Refer to SDS section 2 for GHS hazard classes applicable for this product.

CERCLA RQ	CAS Number	Chemical name

100 LBS	123-72-8	butyraldehyde

State regulations

State RTK	CAS Number	Chemical name
NJ	123-05-7	2-ethylhexanal
	123-72-8	butyraldehyde
PA	123-05-7	2-ethylhexanal
	123-72-8	butyraldehyde
	645-62-5	2-ethylhex-2-enal

NFPA Hazard codes:
Health: 3 Fire: 2 Reactivity: 0 Special:

16. Other Information

SDS Prepared by:
BASF NA Product Regulations
SDS Prepared on: 2019/07/11

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